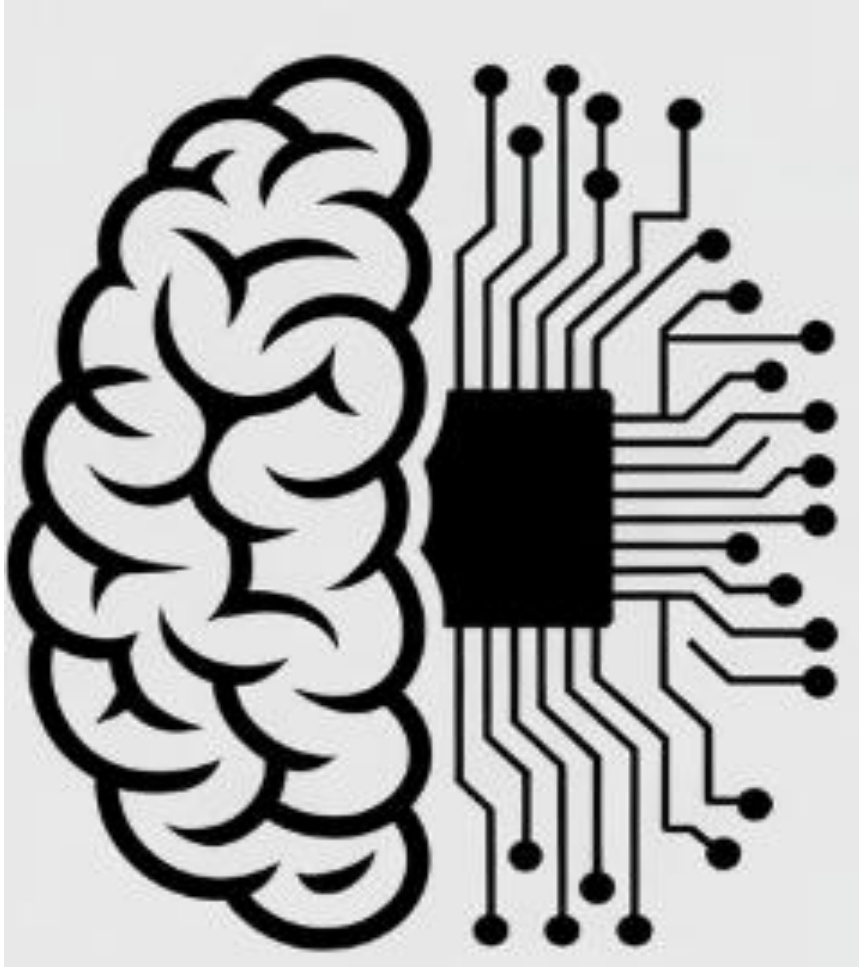
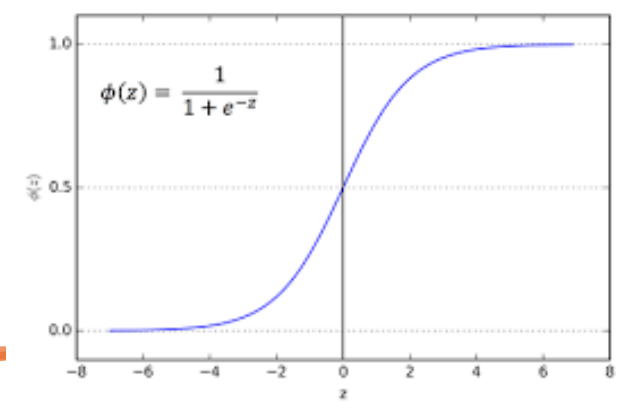




Neural Network Back Propagation

Dr. Sobia Tariq Javed

Sigmoid function



- $\sigma(x) = \frac{1}{1+e^{-x}}$
- $\frac{d\sigma(x)}{dx} = \frac{1}{1+e^{-x}}$
- $\frac{d\sigma(x)}{dx} = (1 + e^{-x})^{-1}$
- $\sigma'(x) = (-1)(1 + e^{-x})^{-2}(-1)(e^{-x})$
 $= \frac{e^{-x}}{(1+e^{-x})^2}$
 $= \left[\frac{1}{1+e^{-x}} \right] \left[\frac{e^{-x}}{1+e^{-x}} \right]$
 $= \left[\frac{1}{1+e^{-x}} \right] \left[\frac{-1+1+e^{-x}}{1+e^{-x}} \right]$

- $\sigma'(x) = \left[\frac{1}{1+e^{-x}} \right] \left[1 - \frac{1}{1+e^{-x}} \right]$
- Since $\sigma(x) = \frac{1}{1+e^{-x}}$
- $\sigma'(x) = \sigma(x)[1 - \sigma(x)]$

Delta Rule

- Delta rule **updates weights** proportionally to the **error**, helping the model learn by minimizing the difference between **predicted and target outputs**.
- $W_i^{\text{new}} = W_i^{\text{old}} + \Delta W_i$
- $\Delta W_i = \text{learning rate} * (\text{Actual output} - \text{predicted output}) * x_i$
- $W_i^{\text{new}} = W_i^{\text{old}} + \eta(y - \hat{y})x_i$
- It can only learn **linearly separable patterns**.
- It is used with **linear neurons (Perceptron) or (single-layer networks)**.

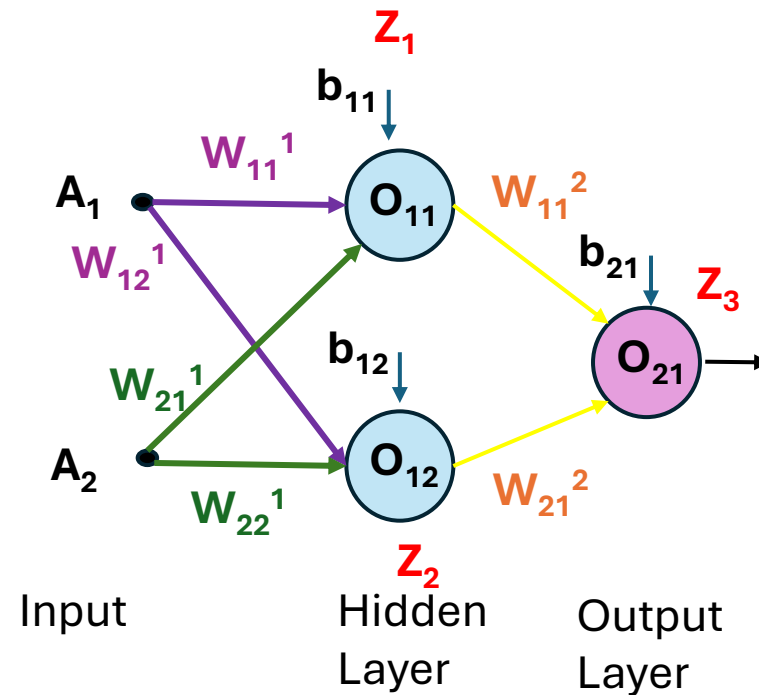
Gradient Descent

- Gradient descent is an iterative method that **updates parameters** in the **direction of the negative gradient** to minimize the **loss function**.
- $W_i^{\text{new}} = W_i^{\text{old}} - \Delta W_i$
- $\Delta W_i = \text{learning rate} * \text{gradient of Error/loss function w.r.t weight}$
- $W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial E}{\partial W_i}$
- Its general optimization method (used in many algorithms).

Multi-layer NN

$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

	A ₁	A ₂	Labels
X ₁	X ₁₁	X ₁₂	L ₁
X ₂	X ₂₁	X ₂₂	L ₂
X ₃	X ₃₁	X ₃₂	L ₃
⋮	⋮	⋮	⋮
X _m	X _{m1}	X _{m2}	L _m



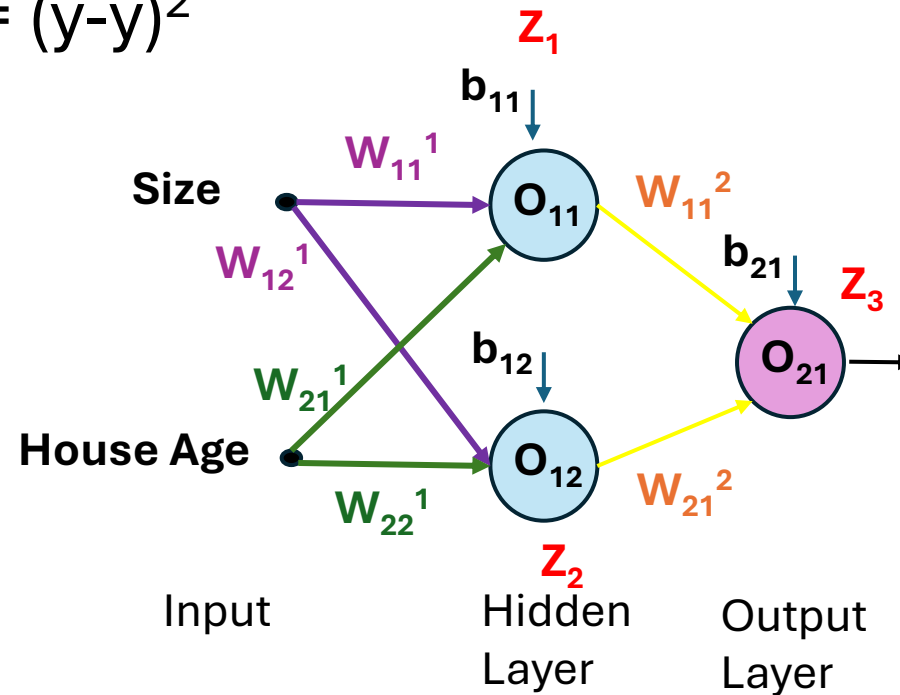
9 trainable parameters

Multi-layer NN- Regression

$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

- Loss Function = $L = \text{MSE} = (y - \hat{y})^2$

	Size	House Age	Price
X_1	5	13.2	90
X_2	10	5	140
X_3	20	25	200
$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_m	12	45	145



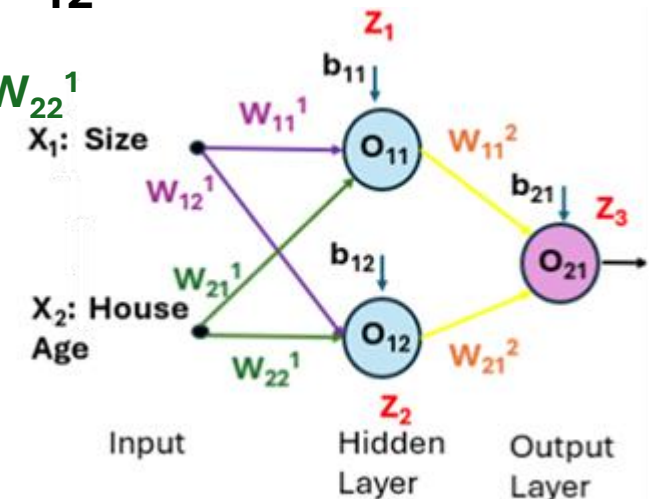
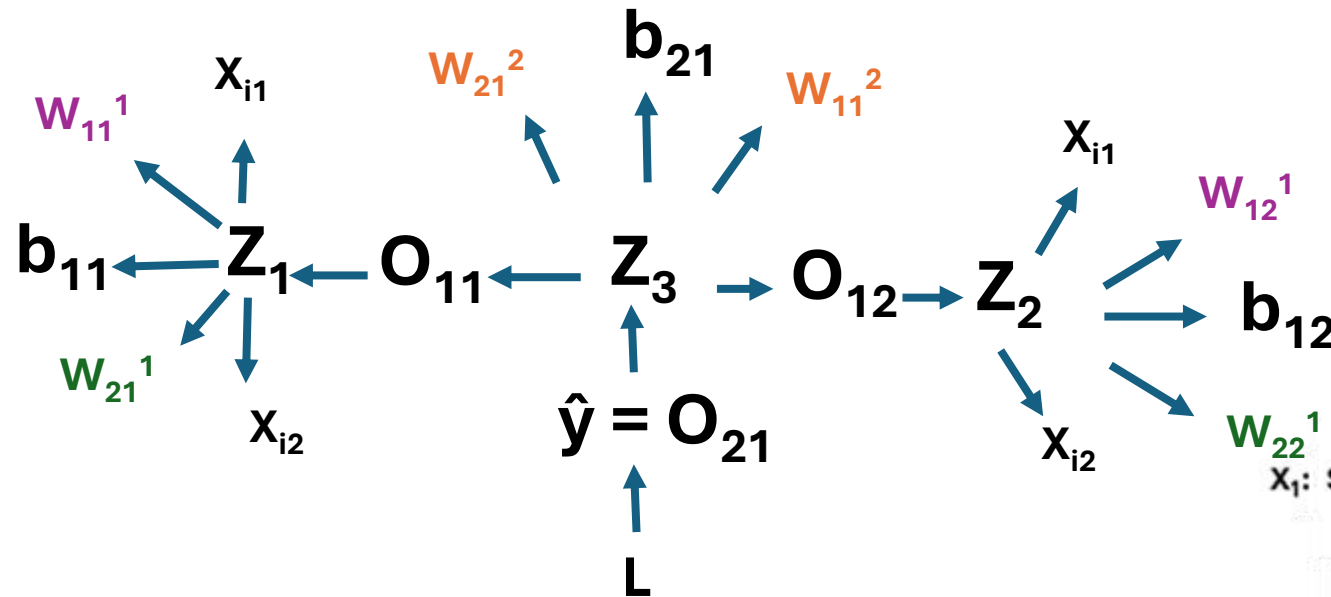
9 trainable parameters

Multi-layer NN- Regression

$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

- Loss Function = $L = \text{MSE} = (y - \hat{y})^2$

	Size	House Age	Price
X_1	5	13.2	90
X_2	10	5	140
X_3	20	25	200
$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_m	12	45	145



Multi-layer NN- Regression- Layer 2

- $L = (y - \hat{y})^2$

- $\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial W_{11}^2}$ $\frac{\partial L}{\partial W_{11}^2} = ?$, $\frac{\partial L}{\partial W_{21}^2} = ?$, $\frac{\partial L}{\partial b_{21}} = ?$

- $\frac{\partial L}{\partial \hat{y}} = \frac{\partial (y - \hat{y})^2}{\partial \hat{y}} = 2(y - \hat{y})(-1)$

- $\frac{\partial \hat{y}}{\partial Z_3} = 1$

- $\frac{\partial Z_3}{\partial W_{11}^2} = O_{11}$

- $\frac{\partial L}{\partial W_{11}^2} = -2(y - \hat{y}) O_{11}$

- $\frac{\partial L}{\partial W_{21}^2} = -2(y - \hat{y}) O_{12}$

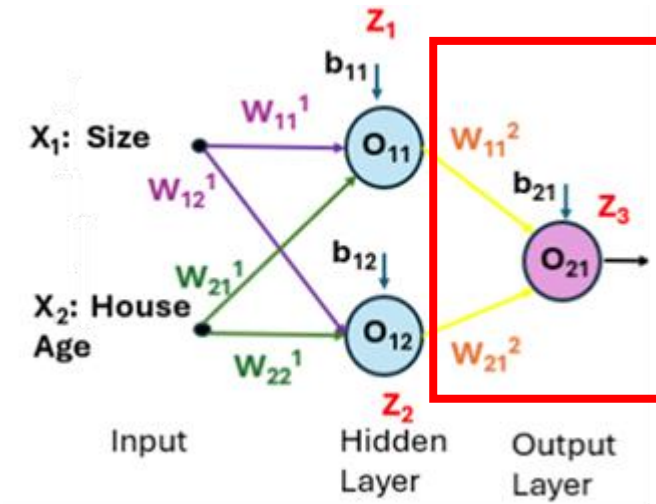
- $\frac{\partial L}{\partial b_{21}} = -2(y - \hat{y})$

$$\hat{y} = O_{21} = Z_3$$

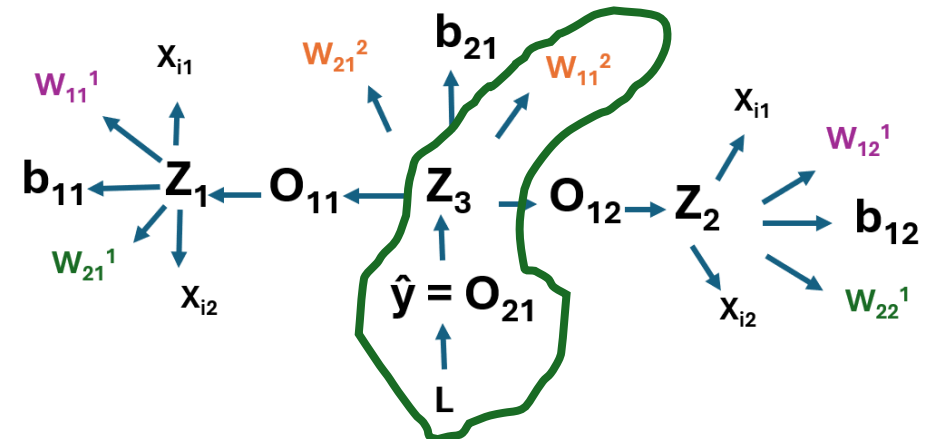
$$Z_3 = O_{11}W_{11}^2 + O_{12}W_{21}^2 + b_{21}$$

$$\frac{\partial \hat{y}}{\partial W_{21}^2} = O_{12}$$

$$\frac{\partial \hat{y}}{\partial W_{21}^2} = 1$$



$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$



Multi-layer NN- Regression-Layer 2

- $L = (y - \hat{y})^2$

- $\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial W_{11}^2}$ $\frac{\partial L}{\partial W_{11}^2} = ?$, $\frac{\partial L}{\partial W_{21}^2} = ?$, $\frac{\partial L}{\partial b_{21}} = ?$

- $\frac{\partial L}{\partial W_{11}^2} = -2(y - \hat{y}) O_{11}$

- $\frac{\partial L}{\partial W_{21}^2} = -2(y - \hat{y}) O_{12}$

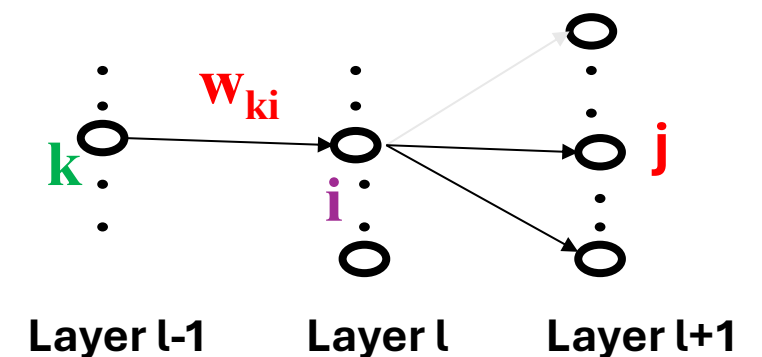
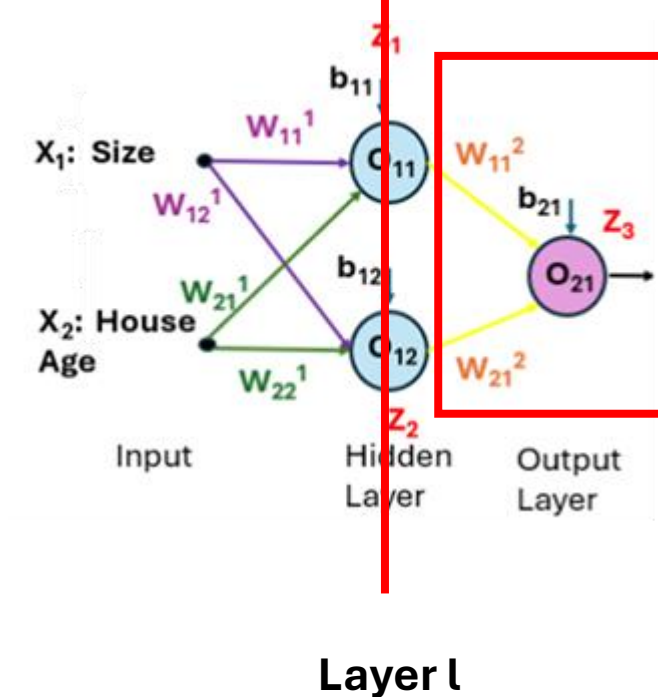
- $\frac{\partial L}{\partial b_{21}} = -2(y - \hat{y})$

- General $\frac{\partial L}{\partial W_{ij}^{(l+1)}} = \frac{\partial L}{\partial \hat{y}_j} \cdot \frac{\partial \hat{y}_j}{\partial Z_{(l+1)j}} \cdot \frac{\partial Z_{(l+1)j}}{\partial W_{ij}^{(l+1)}}$

- $\frac{\partial L}{\partial W_{11}^{(l+1)}} = c [-2((y - \hat{y})) \cdot \sigma'(Z_{(l+1)j}) \cdot X_{li}]$

Let $X_{li} = O_{1i}$, $i \in \{1, 2\}$

Let $\sigma'(Z_{(l+1)j}) = \frac{\partial \hat{y}}{\partial Z_3}$, $Z_{(l+1)j} = Z_3$

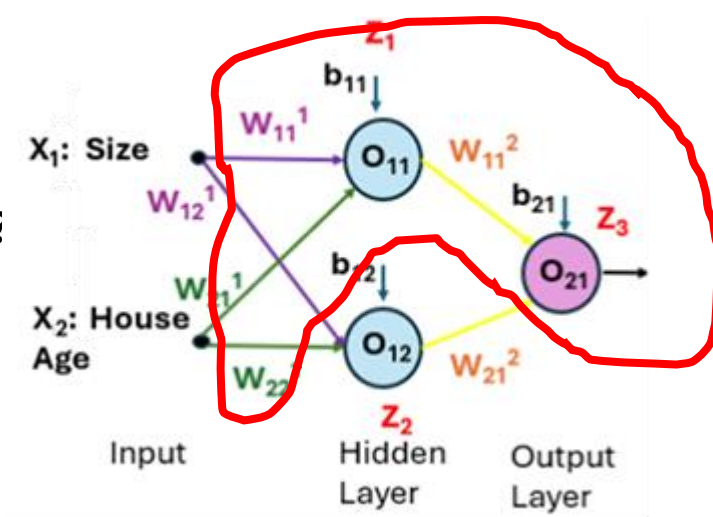


Multi-layer NN- Regression-Layer 1

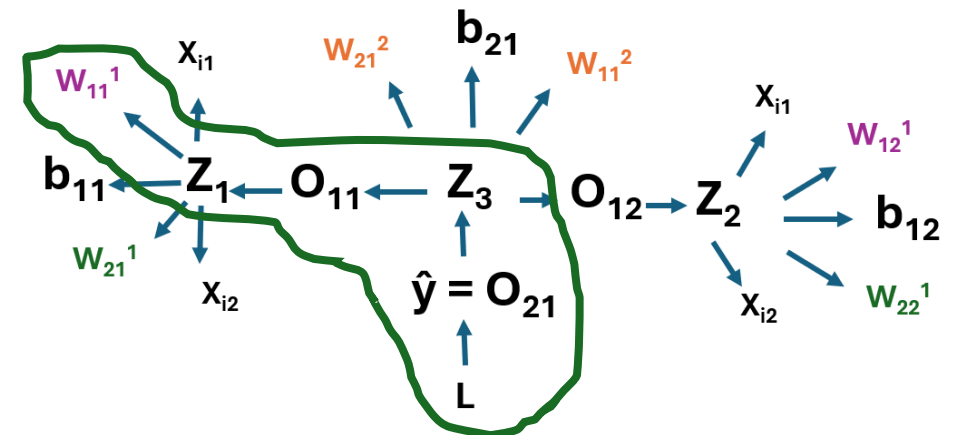
- $L = (y - \hat{y})^2$

$$\frac{\partial L}{\partial W_{11}^1} = ? , \frac{\partial L}{\partial W_{21}^1} = ? , \frac{\partial L}{\partial b_{11}} = ?$$

- $$\frac{\partial L}{\partial W_{11}^1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial O_{11}} \cdot \frac{\partial O_{11}}{\partial Z_1^1} \cdot \frac{\partial Z_1^1}{\partial W_{11}^1}$$



$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

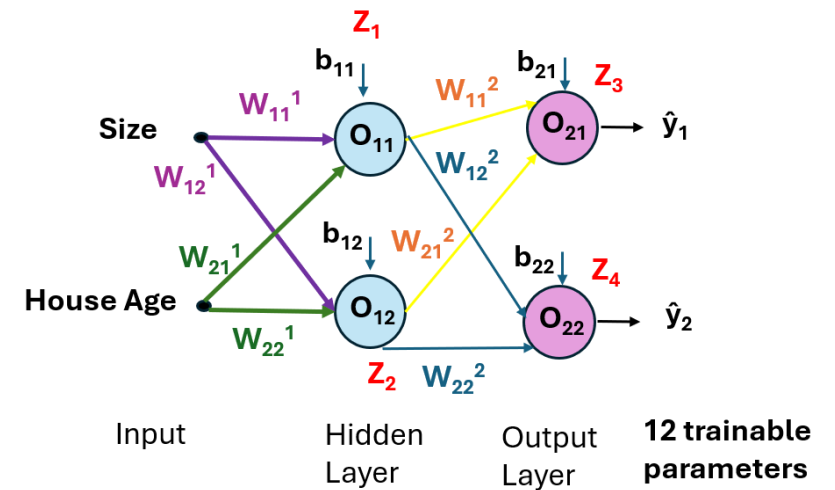
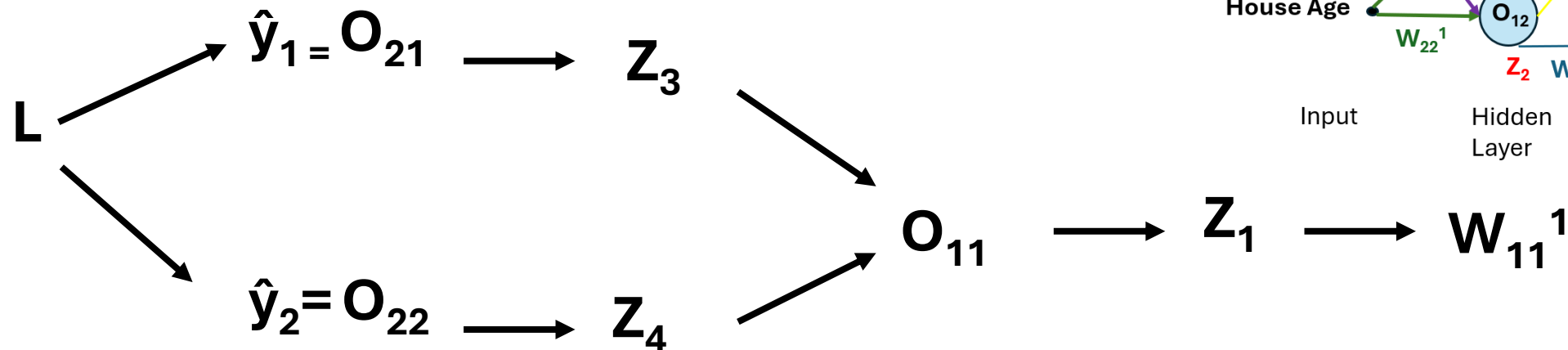


Multi-layer NN- Regression-Layer 1

- What if we have 2 neurons in output layer instead of one.

- $L = (y - \hat{y})^2$

- $$\frac{\partial L}{\partial W_{11}^1} = \left\{ \frac{\partial L}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial O_{11}} + \frac{\partial L}{\partial \hat{y}_2} \cdot \frac{\partial \hat{y}_2}{\partial Z_4} \cdot \frac{\partial Z_4}{\partial O_{11}} \right\} \cdot \frac{\partial O_{11}}{\partial Z_1^1} \cdot \frac{\partial Z_1^1}{\partial W_{11}^1}$$



Multi-layer NN- Regression-Layer 1

- $L = (y - \hat{y})^2$

$$\frac{\partial L}{\partial W_{11}^1} = ? , \frac{\partial L}{\partial W_{21}^1} = ? , \frac{\partial L}{\partial b_{11}} = ?$$

- $\frac{\partial L}{\partial W_{11}^1} = \boxed{\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial O_{11}} \cdot \frac{\partial O_{11}}{\partial Z_1^1}} \cdot \frac{\partial Z_1^1}{\partial W_{11}^1}$

- $\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} = 2(y - \hat{y})(-1)$

$$\frac{\partial O_{11}}{\partial Z_1^1} = 1$$

- $\frac{\partial Z_3}{\partial O_{11}} = W_{11}^2$

$$Z_3 = O_{11}W_{11}^2 + O_{12}W_{21}^2 + b_{21}$$

- $\frac{\partial Z_1^1}{\partial W_{11}^1} = X_{i1}$

$$Z_1 = X_{i1}W_{11}^1 + X_{i2}W_{21}^1 + b_{11}$$

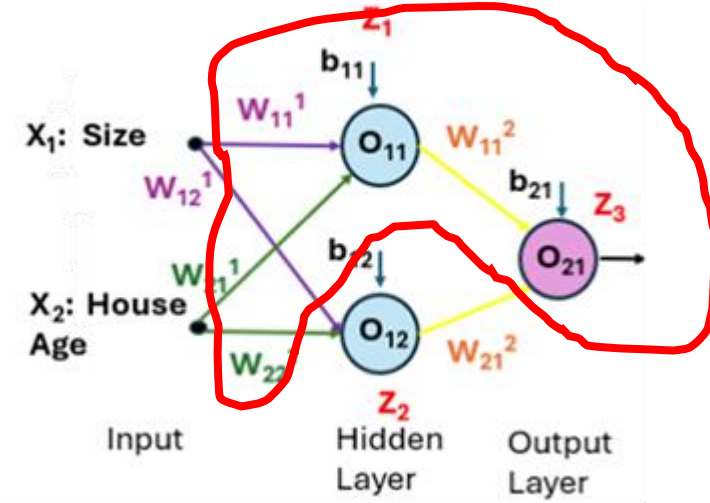
- $\frac{\partial L}{\partial W_{11}^1} = -2(y - \hat{y}) W_{11}^2 X_{i1}$

- $\frac{\partial L}{\partial W_{21}^1} = -2(y - \hat{y}) W_{11}^2 X_{i2}$

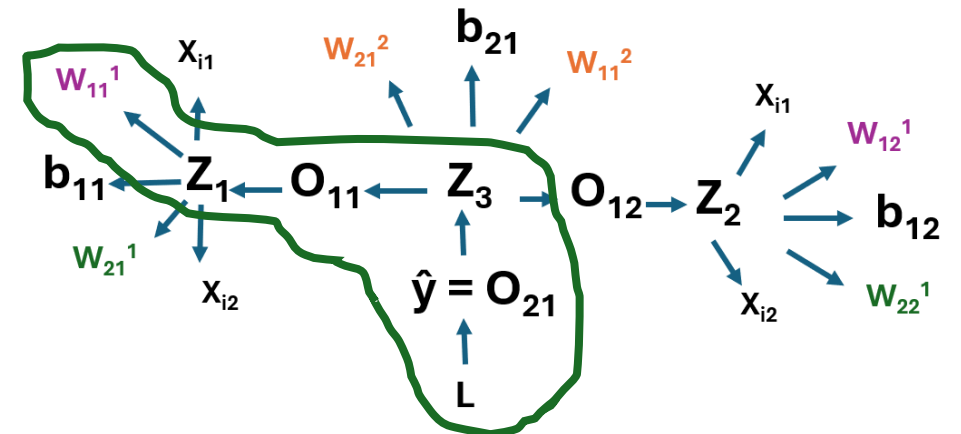
$$\frac{\partial Z_1^1}{\partial W_{21}^1} = X_{i2}$$

- $\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y}) W_{11}^2$

$$\frac{\partial Z_1^1}{\partial b_{11}} = 1$$



$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$



Multi-layer NN- Regression-Layer 1

- $L = (y - \hat{y})^2$

- $\frac{\partial L}{\partial W_{12}^1} = \boxed{\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial O_{12}} \cdot \frac{\partial O_{12}}{\partial Z_2^1}} \cdot \frac{\partial Z_2^1}{\partial W_{12}^1}$

- $\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} = 2(y - \hat{y})(-1)$

$$\frac{\partial O_{12}}{\partial Z_2^1} = 1$$

- $\frac{\partial Z_3}{\partial O_{12}} = W_{21}^2$

$$Z_3 = O_{11}W_{11}^2 + O_{12}W_{21}^2 + b_{21}$$

- $\frac{\partial Z_2^1}{\partial W_{12}^1} = X_{i1}$

$$O_{12} = X_{i1}W_{12}^1 + X_{i2}W_{22}^1 + b_{12}$$

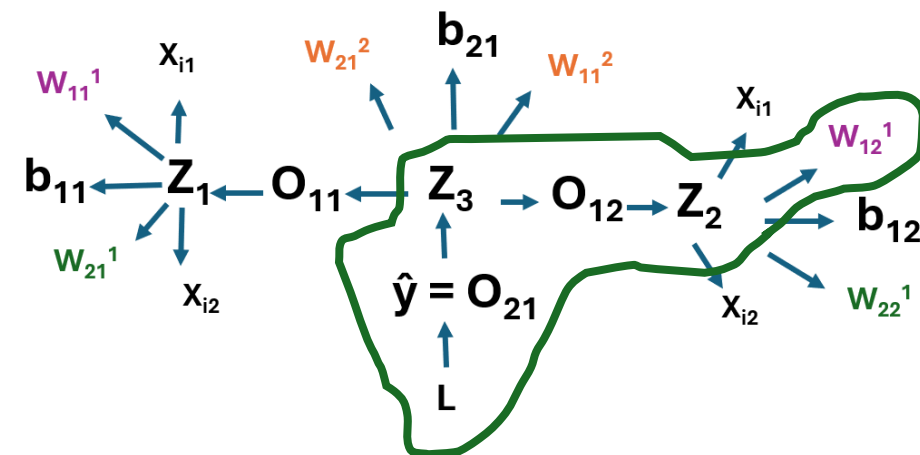
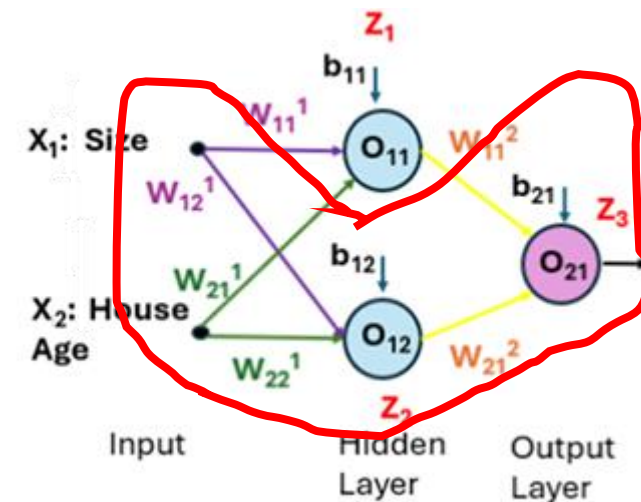
- $\frac{\partial L}{\partial W_{12}^1} = -2(y - \hat{y})W_{21}^2X_{i1}$

- $\frac{\partial L}{\partial W_{22}^1} = -2(y - \hat{y})W_{21}^2X_{i2}$

$$\frac{\partial Z_2^1}{\partial W_{22}^1} = X_{i2}$$

- $\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y})W_{12}^2$

$$\frac{\partial Z_2^1}{\partial b_{12}} = 1$$



Multi-layer NN- Regression-Layer 1

- $L = (y - \hat{y})^2$

- $\frac{\partial L}{\partial W_{12}^1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial O_{12}} \cdot \frac{\partial O_{12}}{\partial Z_2^1} \cdot \frac{\partial Z_2^1}{\partial W_{12}^1}$

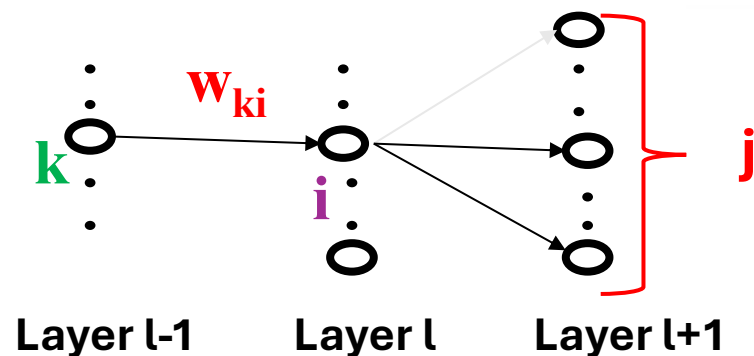
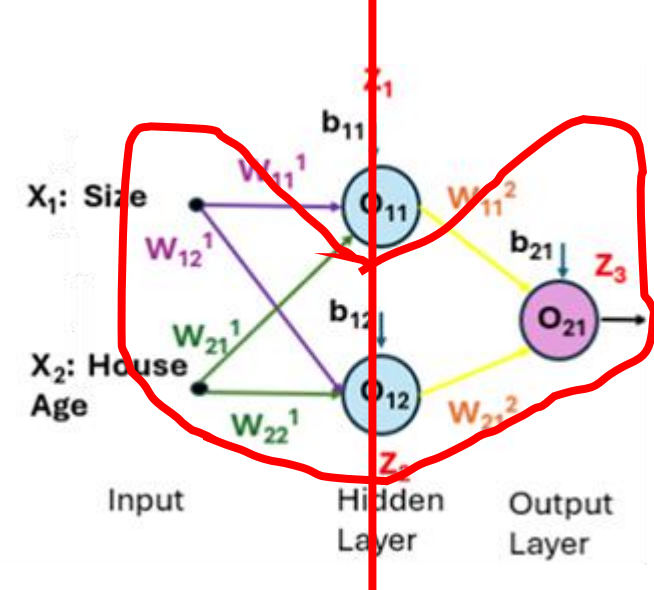
- $\frac{\partial L}{\partial W_{12}^1} = -2(y - \hat{y}) W_{21}^2 X_{i1}$

- $\frac{\partial L}{\partial W_{22}^1} = -2(y - \hat{y}) W_{21}^2 X_{i2}$

- $\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y}) W_{12}^2$

- General $\frac{\partial L}{\partial W_{ki}^{(l-1)}} = \sum_j \left(\frac{\partial L}{\partial \hat{y}_j} \cdot \frac{\partial \hat{y}_j}{\partial Z_{(l+1)j}} \cdot \frac{\partial Z_{(l+1)j}}{\partial O_{li}} \right) \cdot \frac{\partial O_{li}}{\partial Z_{li}^1} \cdot \frac{\partial Z_{li}^1}{\partial W_{ki}^1}$

- $\frac{\partial L}{\partial W_{ki}^{(l-1)}} = -c [-2 \sum_j (y_j - \hat{y}_j) \cdot \sigma'(Z_{(l+1)j}) \cdot W_{ij} \cdot \sigma'(Z_{li}) \cdot X_k]$



$$W_i^{\text{new}} = W_i^{\text{old}} - \eta^* \frac{\partial L}{\partial W_i}$$

Algorithm

- Epochs = 5

for i in range (epochs)

for j in range (X. Shape[0] → rows):

Select row

Predict (using forward pass)

Calculate Loss (using loss function → MSE)

Compute ΔW and Δb .

Update weights & bias using GD

$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

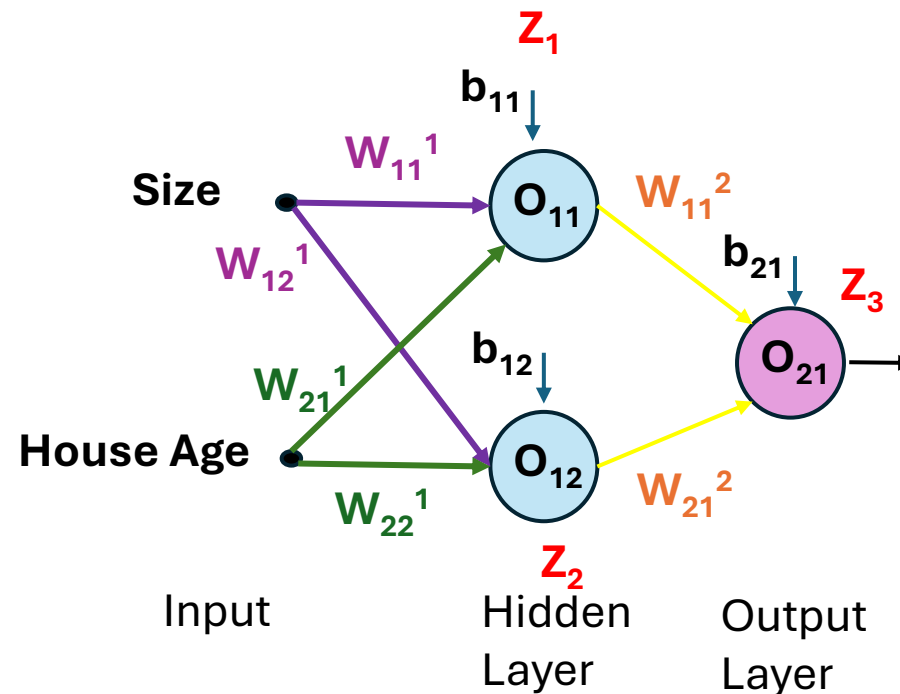
Calculate avg loss for the Epoch

$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

Multi-layer NN- Classification

- Loss Function = L = Binary Cross-Entropy = $-y\log(\hat{y}) - (1-y)\log(1 - \hat{y})$
- Activation Function = Sigmoid

	Size	House Age	Rebuild
x_1	5	13.2	0
x_2	10	5	0
x_3	20	25	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	12	45	1



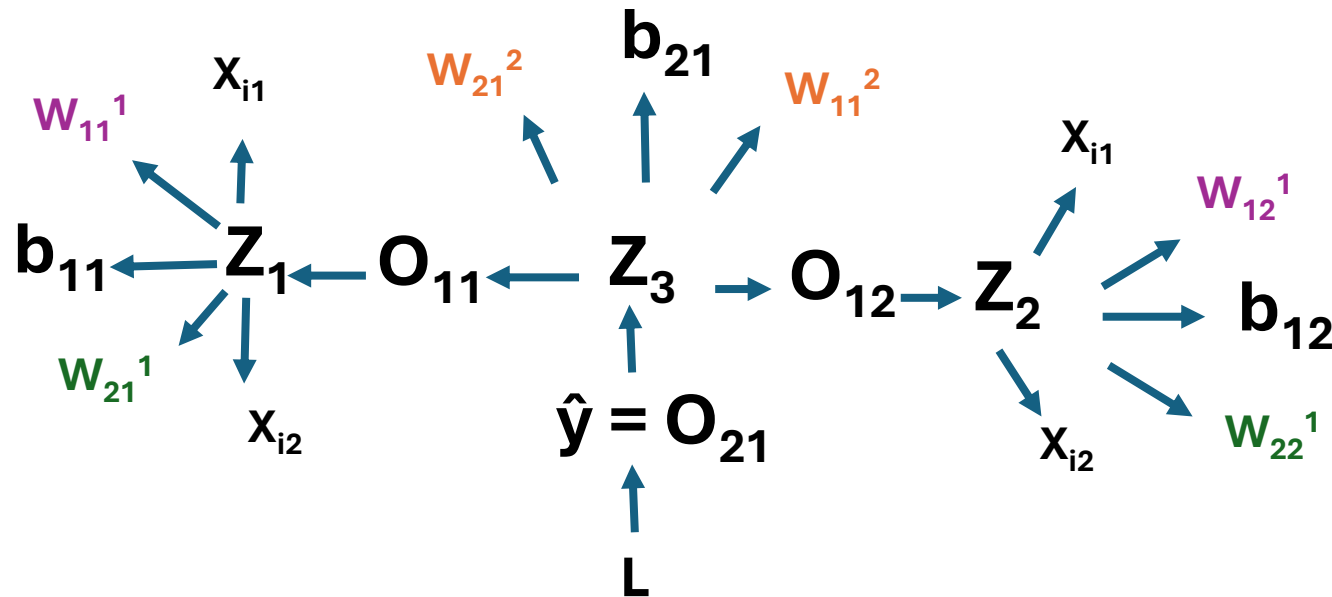
9 trainable parameters

$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$

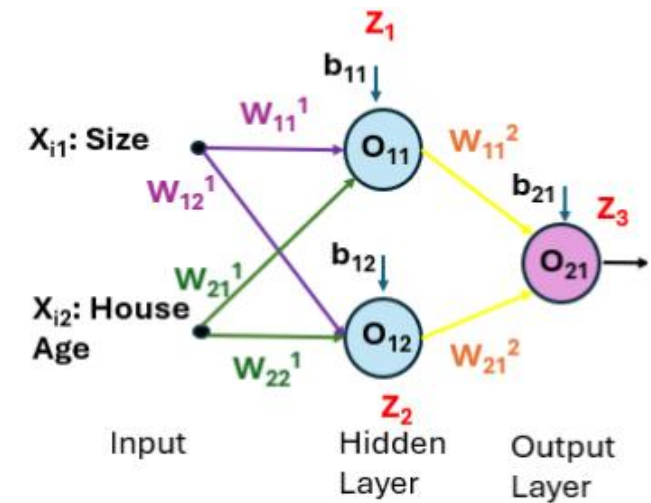
Multi-layer NN- Classification

- Loss Function = L = Binary Cross-Entropy = $-y\log(\hat{y}) - (1-y)\log(1 - \hat{y})$
- Activation Function = Sigmoid

	Size	House Age	Rebuild
X_1	5	13.2	0
X_2	10	5	0
X_3	20	25	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_m	12	45	1



9 trainable parameters



Multi-layer NN- Classification-Layer 2

- $L = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$

- $\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial W_{11}^2}$

- $\frac{\partial L}{\partial \hat{y}} = \frac{\partial (-y\log(\hat{y}) - (1-y)\log(1-\hat{y}))}{\partial \hat{y}}$

$$= \frac{-y}{\hat{y}} - \frac{(1-y)}{(1-\hat{y})(-1)}$$

$$= \frac{-y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})}$$

$$= \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{(1-\hat{y})(\hat{y})}$$

$$= \frac{-y + (y\hat{y}) + \hat{y} - (\hat{y}y)}{(1-\hat{y})(\hat{y})}$$

$$= \frac{\hat{y} - y}{(1-\hat{y})(\hat{y})} = \frac{-(y-\hat{y})}{(1-\hat{y})(\hat{y})}$$

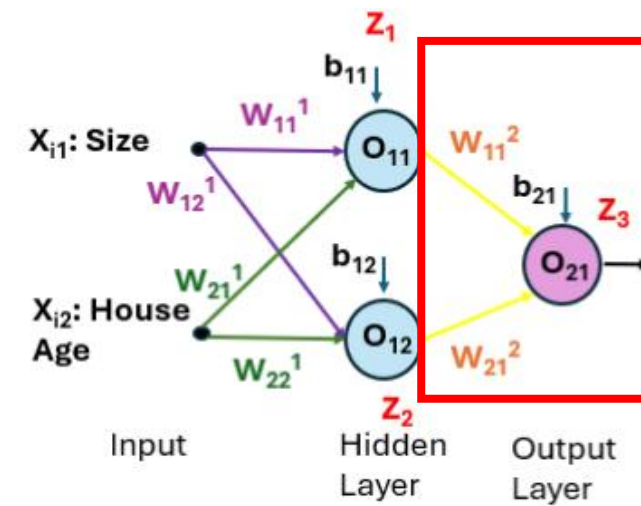
$$\frac{\partial L}{\partial W_{11}^2} = ?, \quad \frac{\partial L}{\partial W_{21}^2} = ?, \quad \frac{\partial L}{\partial b_{21}} = ?$$

$$\frac{\partial \log(\hat{y})}{\partial \hat{y}} = \frac{1}{\hat{y}}$$

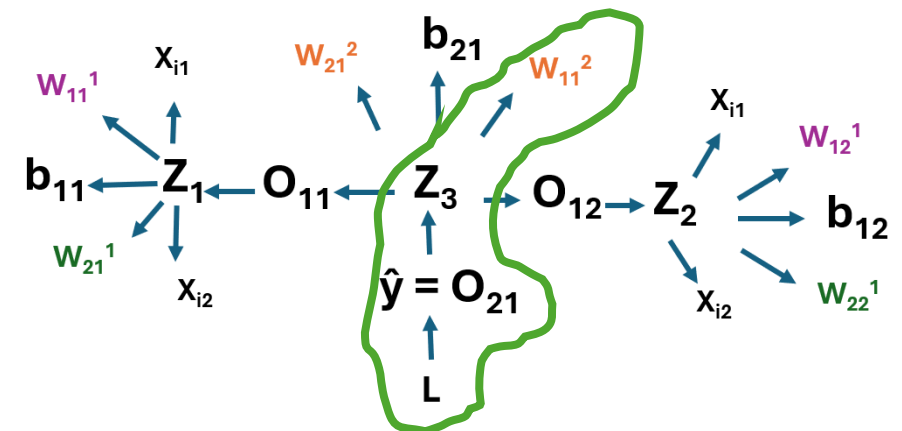
H.W:

Compute $\frac{\partial L}{\partial W_{11}^2}$ using

$$\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial W_{11}^2}$$



$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$



Multi-layer NN- Classification-Layer 2

- $L = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$

- $\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z_3} \cdot \frac{\partial Z_3}{\partial W_{11}^2}$

- $\frac{\partial L}{\partial \hat{y}} = \frac{-(y-\hat{y})}{(1-\hat{y})(\hat{y})}$

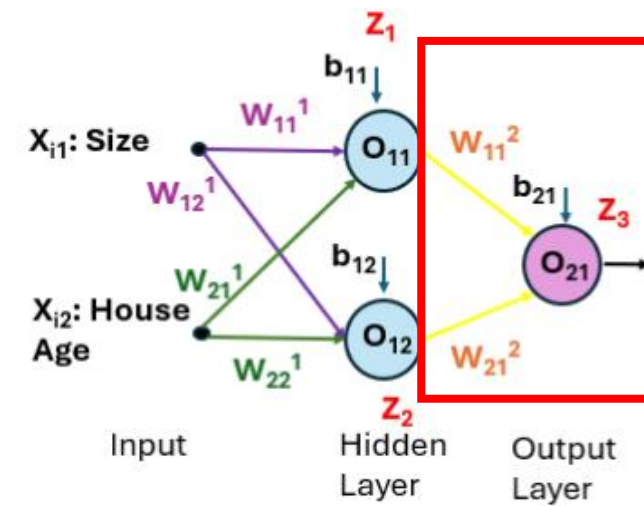
- Using General Formula

$$\frac{\partial L}{\partial W_{11}^{(l+1)}} = \left[\frac{\partial L}{\partial \hat{y}} \cdot \sigma'(Z_{(l+1)j}) \cdot X_{li} \right]$$

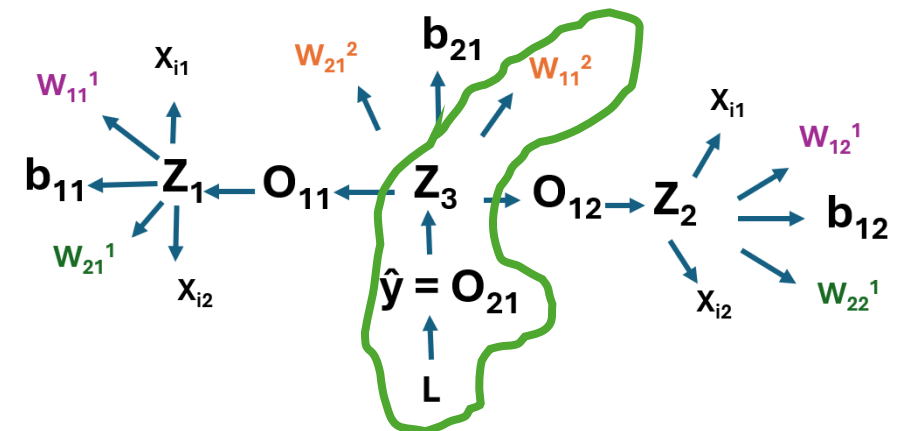
- $\frac{\partial L}{\partial W_{11}^2} = \frac{-(y-\hat{y})}{(1-\hat{y})(\hat{y})} \cdot \sigma(Z_3)(1-\sigma(Z_3)) \cdot O_{11}$

- $\frac{\partial L}{\partial W_{11}^2} = \frac{-(y-\hat{y})}{(1-\hat{y})(\hat{y})} \cdot \hat{y}(1-\hat{y}) \cdot O_{11} = -(y-\hat{y}) O_{11}$

$$\frac{\partial L}{\partial W_{11}^2} = ?, \quad \frac{\partial L}{\partial W_{21}^2} = ?, \quad \frac{\partial L}{\partial b_{21}} = ?$$



$$W_i^{\text{new}} = W_i^{\text{old}} - \eta * \frac{\partial L}{\partial W_i}$$



Multi-layer NN- Classification-Layer 2

- $L = -y\log(\hat{y}) - (1-y)\log(1 - \hat{y})$

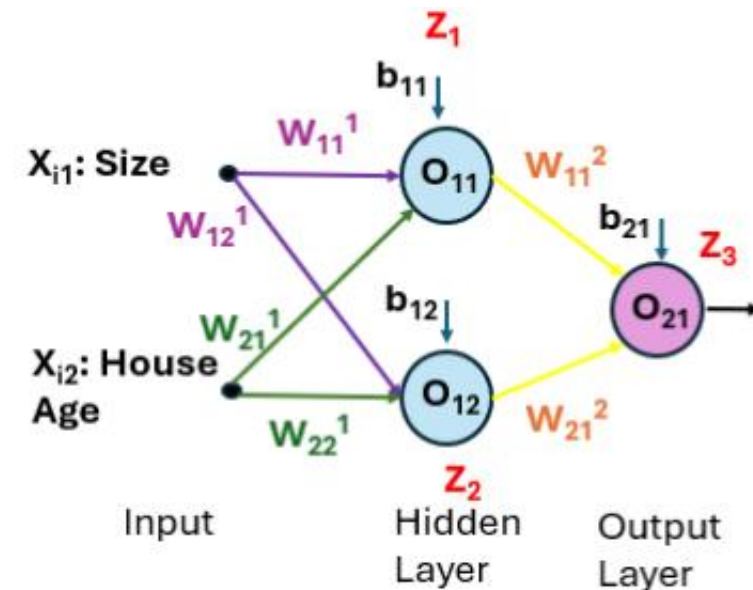
- Using General Formula

$$\frac{\partial L}{\partial W_{11}^{(l+1)}} = \left[\frac{\partial L}{\partial \hat{y}} \cdot \sigma'(Z_{(l+1)j}) \cdot X_{li} \right]$$

- $\frac{\partial L}{\partial W_{11}^2} = -(y - \hat{y}) \mathbf{O}_{11}$

- $\frac{\partial L}{\partial W_{21}^2} = -(y - \hat{y}) \mathbf{O}_{12}$

- $\frac{\partial L}{\partial b_{21}} = -(y - \hat{y})$



Multi-layer NN- Classification-Layer 1

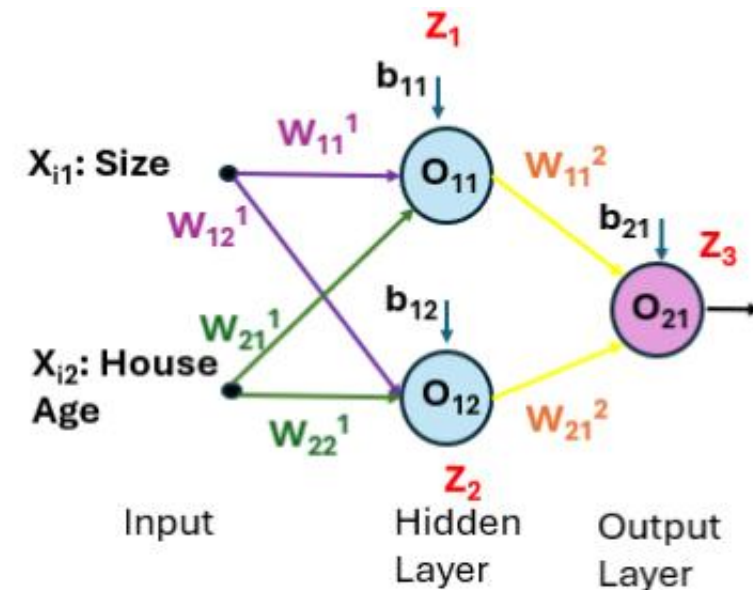
- $L = -y\log(\hat{y}) - (1-y)\log(1 - \hat{y})$

- **Using General Formula** $\frac{\partial L}{\partial W_{ki}^{(l-1)}} = \sum_j \left(\frac{\partial L}{\partial \hat{y}_j} \cdot \frac{\partial \hat{y}_j}{\partial Z_{(l+1)j}} \cdot \frac{\partial Z_{(l+1)j}}{\partial O_{li}} \right) \cdot \frac{\partial O_{li}}{\partial Z_{li}^1} \cdot \frac{\partial Z_{li}^1}{\partial W_{ki}^1}$

- $\frac{\partial L}{\partial W_{11}^1} = \frac{-(y - \hat{y})}{(1 - \hat{y})(\hat{y})} \cdot (\sigma(Z_3)(1 - \sigma(Z_3)) \cdot W_{11}^2 \sigma(Z_1)(1 - \sigma(Z_1)) X_1$
 $= (y - \hat{y}) \cdot W_{11}^2 O_{11}(1 - O_{11}) X_1$

- $\frac{\partial L}{\partial W_{21}^1} = -(y - \hat{y}) \cdot W_{11}^2 O_{11}(1 - O_{11}) X_2$

- $\frac{\partial L}{\partial b_{11}^1} = (y - \hat{y}) \cdot W_{11}^2 O_{11}(1 - O_{11})$



Multi-layer NN- Classification-Layer 1

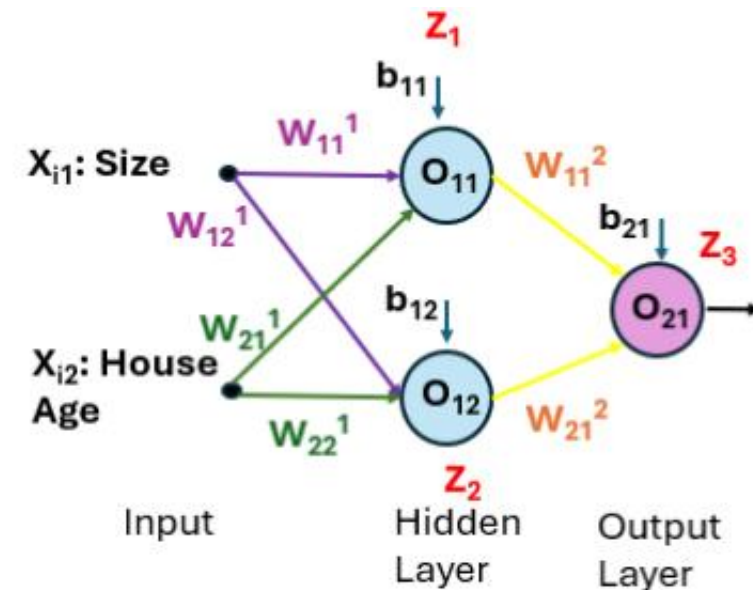
- $L = -y\log(\hat{y}) - (1-y)\log(1 - \hat{y})$

- Using General Formula
$$\frac{\partial L}{\partial W_{ki}^{(l-1)}} = \sum_j \left(\frac{\partial L}{\partial \hat{y}_j} \cdot \frac{\partial \hat{y}_j}{\partial Z_{(l+1)j}} \cdot \frac{\partial Z_{(l+1)j}}{\partial O_{li}} \right) \cdot \frac{\partial O_{li}}{\partial Z_{li}^1} \cdot \frac{\partial Z_{li}^1}{\partial W_{ki}^1}$$

- $\frac{\partial L}{\partial W_{12}^1} = -(y - \hat{y}) \cdot W_{21}^2 O_{12}(1 - O_{12}) X_2$

- $\frac{\partial L}{\partial W_{22}^1} = -(y - \hat{y}) \cdot W_{21}^2 O_{12}(1 - O_{12}) X_2$

- $\frac{\partial L}{\partial b_{11}^1} = -(y - \hat{y}) \cdot W_{21}^2 O_{12}(1 - O_{12})$



Multi-layer NN- Classification-Layer 1

- $L = -y\log(\hat{y}) - (1-y)\log(1 - \hat{y})$

- Using General Formula

- $\frac{\partial L}{\partial W_{11}^1} = -(y - \hat{y}) \mathbf{O}_{11}$

- $\frac{\partial L}{\partial W_{21}^1} = -(y - \hat{y}) \mathbf{O}_{12}$

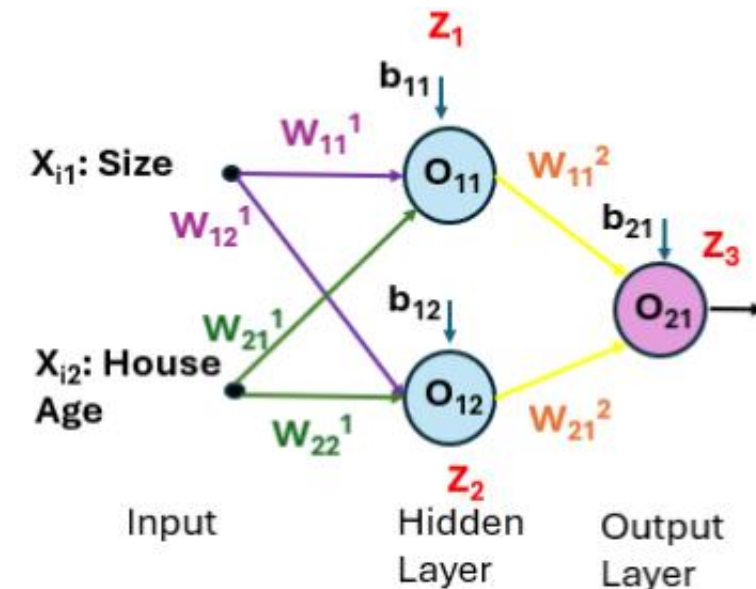
- $\frac{\partial L}{\partial W_{12}^1} = -(y - \hat{y}) \mathbf{O}_{11}$

- $\frac{\partial L}{\partial W_{22}^1} = -(y - \hat{y}) \mathbf{O}_{12}$

- $\frac{\partial L}{\partial b_{11}^1} = -(y - \hat{y})$

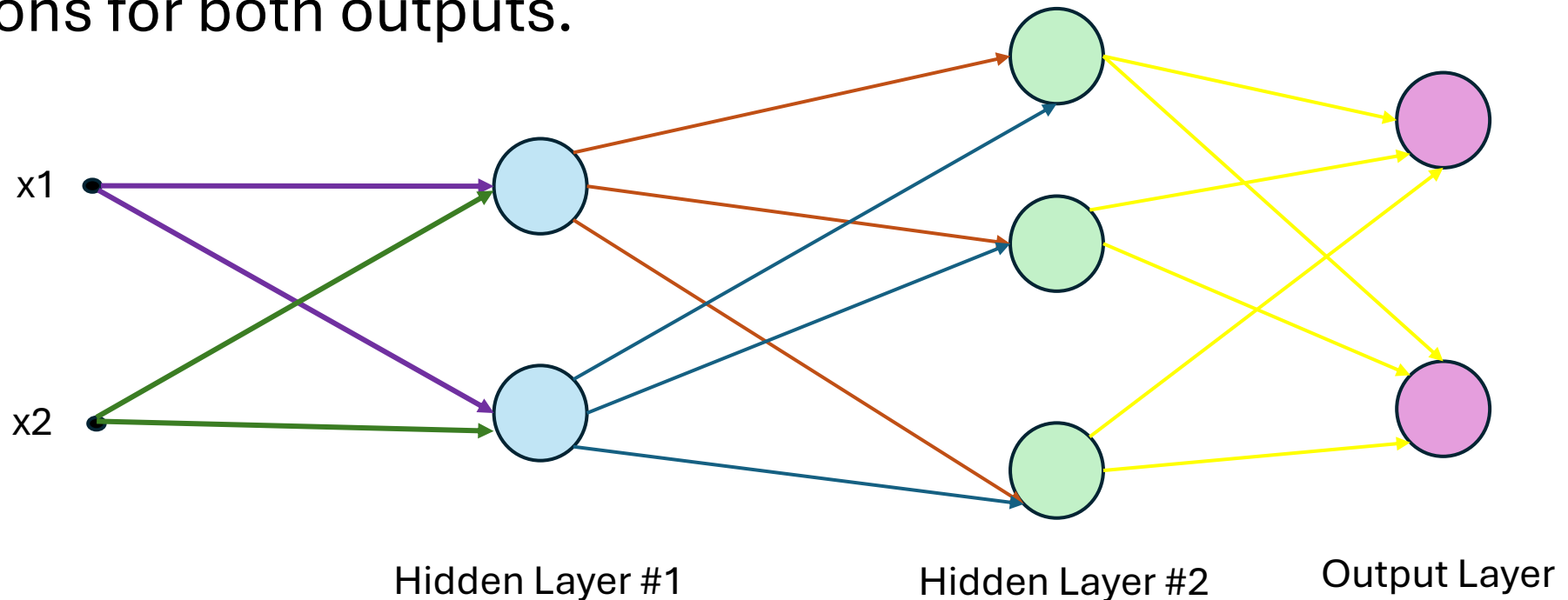
- $\frac{\partial L}{\partial b_{12}^1} = -(y - \hat{y})$

$$\frac{\partial L}{\partial W_{ki}^{(l-1)}} = \sum_j \left(\frac{\partial L}{\partial \hat{y}_j} \cdot \frac{\partial \hat{y}_j}{\partial Z_{(l+1)j}} \cdot \frac{\partial Z_{(l+1)j}}{\partial O_{li}} \right) \cdot \frac{\partial O_{li}}{\partial Z_{li}^1} \cdot \frac{\partial Z_{li}^1}{\partial W_{ki}^1}$$

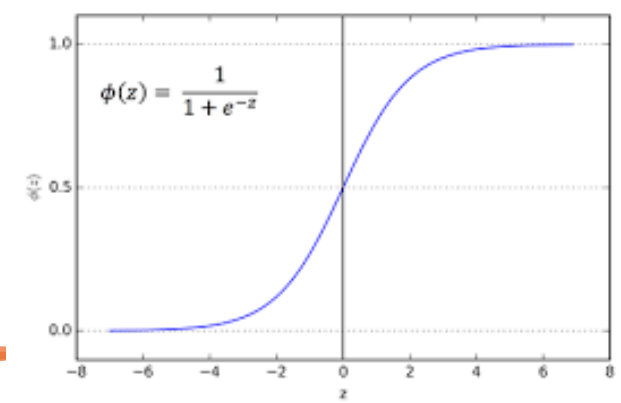


HW

- Apply the Forward + backward pass and label the network clearly according to the conventions covered in class. Compute the equations for both outputs.



Sigmoid function



- $\sigma(x) = \frac{1}{1+e^{-x}}$
- $\frac{d\sigma(x)}{dx} = \frac{1}{1+e^{-x}}$
- $\frac{d\sigma(x)}{dx} = (1 + e^{-x})^{-1}$
- $\sigma'(x) = (-1)(1 + e^{-x})^{-2}(-1)(e^{-x})$
 $= \frac{e^{-x}}{(1+e^{-x})^2}$
 $= \left[\frac{1}{1+e^{-x}} \right] \left[\frac{e^{-x}}{1+e^{-x}} \right]$
 $= \left[\frac{1}{1+e^{-x}} \right] \left[\frac{-1+1+e^{-x}}{1+e^{-x}} \right]$

- $\sigma'(x) = \left[\frac{1}{1+e^{-x}} \right] \left[1 - \frac{1}{1+e^{-x}} \right]$
- Since $\sigma(x) = \frac{1}{1+e^{-x}}$
- $\sigma'(x) = \sigma(x)[1 - \sigma(x)]$

Reference

- Chapter 4 of Tom M. Mitchell “Machine Learning”

text



text



text



text



text



text

